=== UNIT 3: POLYMERS, COMPOSITES & FUNCTIONAL NANOMATERIALS ===

>> 3.1 Classification and Polymerization Mechanisms

Polymers are macromolecules synthesized by covalent linking of repeating monomer units.
* Classification: By origin (natural, synthetic), structure (linear, branched, cross-linked), thermal behavior (Thermoplastics: soften reversibly upon heating; Thermosets: cross-link permanently, decompose without melting).
* Polymerization Mechanisms:
1. Addition (Chain-Growth) Polymerization: Monomers with double bonds link without byproduct: Initiation (free radical from BPO, cationic, or anionic), Propagation, Termination.
2. Condensation (Step-Growth) Polymerization: Polyfunctional monomers react with elimination of small molecules (H_2O, HCl, NH_3).
* Coordination Polymerization: Ziegler-Natta Catalysts ($TiCl_4$ + $Al(C_2H_5)_3$) enable stereospecific polymerization, yielding high-density linear polyethylene (HDPE) with high crystallinity and melting point.

>> 3.2 Engineering Thermoplastics and Thermosets

* Bakelite (Phenol-Formaldehyde Resin): Thermosetting polymer. Phenol and formaldehyde react under acid/base catalysis to form ortho/para-hydroxybenzyl alcohols, condensing into linear Novolac (varnishes). Heating Novolac with hexamethylenetetramine (hexa) introduces methylene cross-links, forming hard, rigid 3D infusible Bakelite (electrical insulators, plugs).
* Nylon-6,6: Polyamide synthesized by condensation of Adipic Acid and Hexamethylenediamine at 280 deg C with elimination of water. Features extensive intermolecular hydrogen bonding between amide groups (-CO-NH-), conferring high tensile strength, elasticity, and abrasion resistance.
* Teflon (Polytetrafluoroethylene, PTFE): Synthesized by free-radical emulsion polymerization of tetrafluoroethylene ($F_2C=CF_2$) initiated by ammonium persulfate. High C-F bond energy (485 kJ/mol) confers chemical inertness, high thermal stability (up to 260 deg C), and ultra-low coefficient of friction (0.04).

>> 3.3 Conducting Polymers and Fiber Composites

* Conducting Polymers: Conjugated polyene backbone with alternating single and double carbon-carbon bonds (e.g., polyacetylene, polyaniline, polypyrrole). Delocalized pi-electrons provide baseline conduction. Electrical conductivity increased up to 10^5 S/cm via Doping: p-doping (oxidation with I_2, removing electrons to form positive polarons/bipolarons) or n-doping (reduction with Na).
* Fiber-Reinforced Plastics (FRP): Structural composites comprising a high-strength

reinforcing fiber (carbon fiber, glass fiber, Kevlar aramid) embedded in a continuous polymer matrix (epoxy, polyester).

>> 3.4 Nanomaterials: Carbon Nanotubes, Quantum Dots & Green Synthesis

* Quantum Confinement: When material dimensions shrink below the exciton Bohr radius (<10 nm), continuous energy bands split into discrete quantized levels; band gap widens inversely with particle size squared: E_g(r) = E_{g,bulk} + h^2 / (8 * m_eff * r^2).
* Carbon Nanotubes (CNTs): Cylindrical graphene cylinders. Single-Walled (SWCNT, 1-2 nm diameter) and Multi-Walled (MWCNT). Possess extraordinary tensile strength (~100 GPa, 100x steel) and ballistic electrical conductivity.
* Green Synthesis of Nanoparticles: Biological synthesis using plant phytochemical extracts (polyphenols, flavonoids, terpenoids) or microorganisms to reduce metal salts (AgNO3 -> Ag0 nanoparticles; HAuCl4 -> Au0 nanoparticles) without toxic surfactants.

[SOLVED PROBLEM 1: Molecular Weight Averages of Polymers]

Problem: A polymer sample consists of three monodisperse fractions:
Fraction 1: 100 molecules of molecular weight 10,000 g/mol.
Fraction 2: 200 molecules of molecular weight 20,000 g/mol.
Fraction 3: 100 molecules of molecular weight 50,000 g/mol.
Calculate: (1) Number-Average Molecular Weight (M_n), (2) Weight-Average Molecular Weight (M_w), (3) Polydispersity Index (PDI).
Solution Step 1: Tabulate N_i and M_i:
Fraction 1: N1 = 100, M1 = 10,000 => N1*M1 = 1,000,000; N1*M1^2 = 1.0 * 10^10.
Fraction 2: N2 = 200, M2 = 20,000 => N2*M2 = 4,000,000; N2*M2^2 = 8.0 * 10^10.
Fraction 3: N3 = 100, M3 = 50,000 => N3*M3 = 5,000,000; N3*M3^2 = 25.0 * 10^10.
Sums: sum N_i = 400; sum (N_i*M_i) = 10,000,000; sum (N_i*M_i^2) = 34.0 * 10^10.
Solution Step 2: Calculate Number-Average Molecular Weight M_n:
M_n = sum (N_i * M_i) / sum N_i = 10,000,000 / 400 = 25,000 g/mol.
Solution Step 3: Calculate Weight-Average Molecular Weight M_w:
M_w = sum (N_i * M_i^2) / sum (N_i * M_i) = 34.0 * 10^10 / 10,000,000 = 34,000 g/mol.
Solution Step 4: Calculate Polydispersity Index (PDI):
PDI = M_w / M_n = 34,000 / 25,000 = 1.36.
Conclusion: Since PDI > 1.0, the polymer is polydisperse with moderate molecular weight distribution.

[SOLVED PROBLEM 2: Degree of Polymerization of Nylon-6,6]

Problem: A sample of Nylon-6,6 has a weight-average molecular weight of M_w = 45,200 g/mol. Calculate: (1) Molecular weight of the repeating structural unit, (2) Degree of Polymerization (DP).
Solution Step 1: Chemical Formula of Nylon-6,6 Repeating Unit:
Monomers: Adipic Acid (HOOC-(CH2)4-COOH) + Hexamethylenediamine (H2N-(CH2)6-NH2).
Condensation eliminates 2 H2O molecules per repeating unit:
Repeating Unit Formula: -[NH-(CH2)6-NH-CO-(CH2)4-CO]-.
Elemental Count: C12 H22 N2 O2.
Solution Step 2: Compute Molecular Weight of Repeating Unit M_0:
Carbon: 12 * 12.011 = 144.13 g/mol.
Hydrogen: 22 * 1.008 = 22.18 g/mol.
Nitrogen: 2 * 14.007 = 28.01 g/mol.
Oxygen: 2 * 15.999 = 32.00 g/mol.
M_0 = 144.13 + 22.18 + 28.01 + 32.00 = 226.32 g/mol.
Solution Step 3: Calculate Degree of Polymerization (DP):
DP = M_w / M_0 = 45,200 / 226.32 = 199.7 ~ 200 repeating units.

[SOLVED PROBLEM 3: Quantum Dot Band Gap and Emission Wavelength Tuning]

Problem: A spherical CdSe quantum dot of radius r = 2.0 nm has bulk band gap E_{g,bulk} = 1.74 eV. Using the Brus quantum confinement equation:
E_g(r) = E_{g,bulk} + [h^2 / (8 * r^2)] * (1/m_e* + 1/m_h*) - [1.8 * e^2 / (4*pi*epsilon*r)].
Given effective mass factor (1/m_e* + 1/m_h*) = 8.5 * 10^30 kg^-1 and ignoring Coulomb correction: Calculate: (1) Effective band gap E_g in eV, (2) Photoluminescence emission wavelength lambda in nanometers.
Solution Step 1: Compute Confinement Energy:
h = 6.626 * 10^-34 J*s; r = 2.0 * 10^-9 m.
Confinement = [(6.626 * 10^-34)^2 / (8 * (2.0 * 10^-9)^2)] * (8.5 * 10^30)
= [4.390 * 10^-67 / (3.2 * 10^-17)] * (8.5 * 10^30) = (1.372 * 10^-50) * (8.5 * 10^30)
= 1.166 * 10^-19 Joules.
Convert to eV (1 eV = 1.602 * 10^-19 J): 1.166 * 10^-19 / 1.602 * 10^-19 = 0.728 eV.
Solution Step 2: Total Effective Band Gap:
E_g(r) = 1.74 eV + 0.728 eV = 2.468 eV.
Solution Step 3: Compute Emission Wavelength lambda:
lambda = (h * c) / E_g = (1240 eV*nm) / 2.468 eV = 502.4 nm (Green emission).
Conclusion: Decreasing quantum dot radius from bulk to 2 nm shifts emission into the green visible spectrum.

[SOLVED PROBLEM 4: Rule of Mixtures for Carbon Fiber Reinforced Epoxy Composite]

Problem: A unidirectional carbon-fiber reinforced epoxy composite contains 60% carbon

fiber by volume (V_f = 0.60, V_m = 0.40). The elastic modulus of carbon fiber is E_f = 240 GPa; elastic modulus of epoxy matrix is E_m = 4.0 GPa. Calculate: (1) Longitudinal modulus E_L (isostrain), (2) Transverse modulus E_T (isostress).
Solution Step 1: Longitudinal Modulus E_L (Rule of Mixtures):
E_L = E_f * V_f + E_m * V_m = (240 GPa * 0.60) + (4.0 GPa * 0.40) = 144.0 + 1.6 = 145.6 GPa.
Solution Step 2: Transverse Modulus E_T (Inverse Rule of Mixtures):
1 / E_T = (V_f / E_f) + (V_m / E_m) = (0.60 / 240) + (0.40 / 4.0) = 0.0025 + 0.1000 = 0.1025 GPa^-1.
E_T = 1 / 0.1025 = 9.76 GPa.
Conclusion: The composite exhibits dramatic mechanical anisotropy: E_L (145.6 GPa) is 15 times stronger than E_T (9.76 GPa).

>> 3.5 University Review & Examination Questions

* Q1: Describe the synthesis, structure, and curing mechanism of Bakelite (Phenol-Formaldehyde resin).
Answer: Synthesis proceeds in two stages: (1) Formation of prepolymers: Phenol and Formaldehyde react in the presence of acid or alkali catalysts. Formaldehyde undergoes electrophilic aromatic substitution on phenol at ortho and para positions, yielding o-hydroxybenzyl alcohol and p-hydroxybenzyl alcohol. (2) Novolac vs. Resol: Acid catalysis with excess phenol yields a linear, soluble, fusible thermoplastic polymer called Novolac. Alkaline catalysis with excess formaldehyde yields branched Resols containing reactive methylol (-CH2OH) groups. (3) Curing to Bakelite: When Novolac is heated to 150-200 deg C with hexamethylenetetramine (hexa) and fillers, hexa decomposes into formaldehyde and ammonia. The liberated formaldehyde forms rigid methylene (-CH2-) bridges cross-linking Novolac chains into a hard, rigid, infusible, insoluble 3D thermosetting polymer network.
* Q2: Explain the mechanism of electrical conduction in conducting polymers and describe the role of p-doping and n-doping.
Answer: Pristine conjugated polymers like polyacetylene possess alternating single and double carbon-carbon bonds with continuous overlap of pi-orbitals along the planar backbone. However, Peierls instability opens a finite band gap (~1.5 eV), rendering them semiconductors. Electrical conductivity is increased by orders of magnitude via chemical or electrochemical doping. p-doping involves partial oxidation using electron acceptors (I2, AsF5, FeCl3). An electron is removed from the pi-system, creating a localized radical cation called a Polaron. Removal of a second electron yields a pair of like charges called a Bipolaron. Under an applied electric field, these bipolarons delocalize and migrate along the conjugated backbone via soliton/bipolaron hopping, achieving metallic conductivities up to 10^5 S/cm. n-doping involves partial reduction using alkali metals (Na, K), injecting electrons into the conduction band.

>> 3.6 Key Takeaways and Revision Points

* Polydispersity index PDI = M_w / M_n measures molecular weight distribution; PDI = 1 for monodisperse samples.
* Bakelite is a cross-linked thermoset formed by phenol-formaldehyde condensation.
* Conducting polymers rely on conjugated pi-systems activated by p-type or n-type redox doping.
* Nanomaterials exhibit quantum confinement when dimensions fall below the exciton Bohr radius.